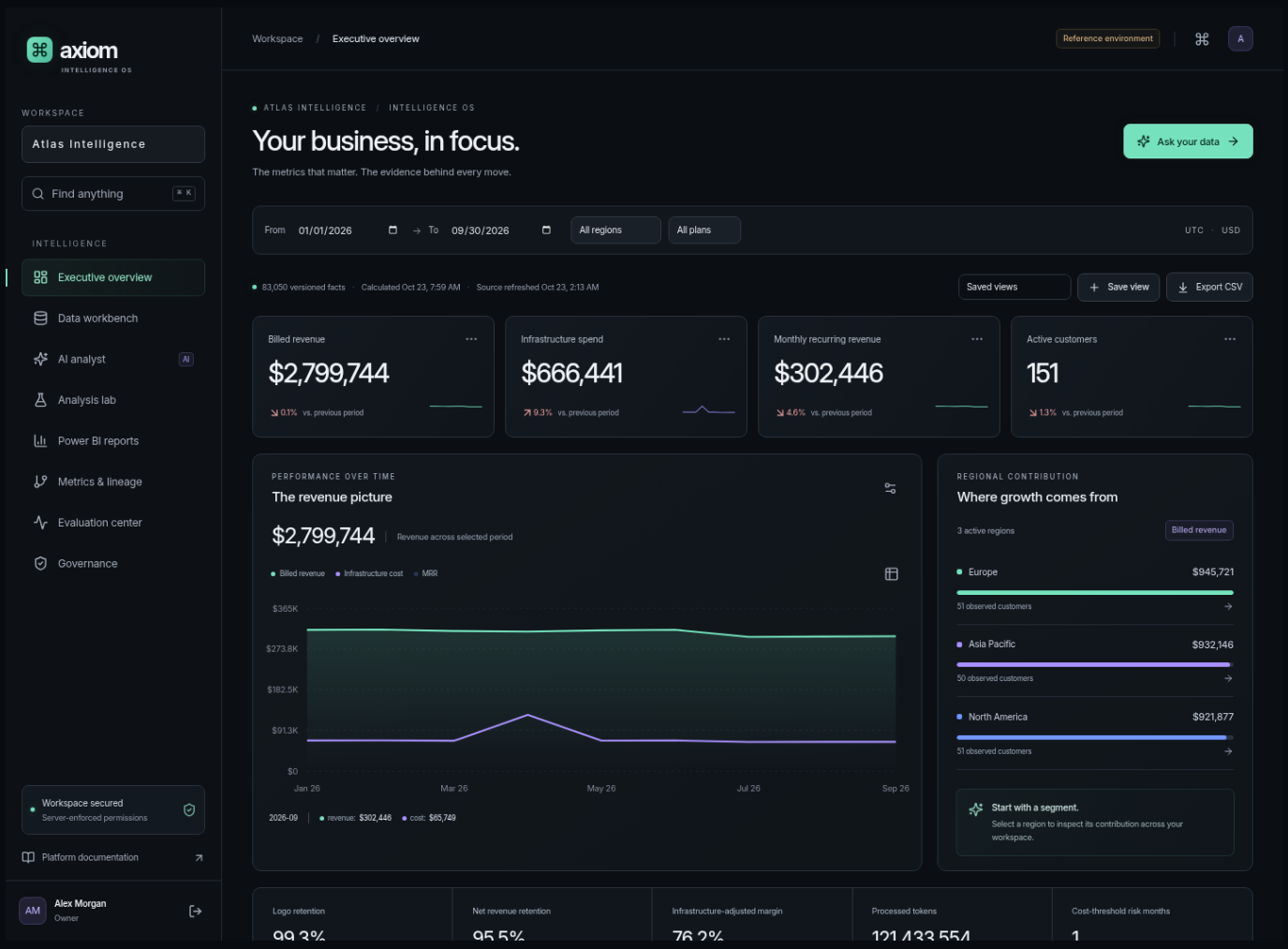
AXIOM / ENGINEERING + PRODUCT HANDOFF

Intelligence OS

A working analytics workspace.
Every answer has a source.

Edition 2.0 · 23 October 2026 · Feature branch review package



249,150 facts	Three organizations, 36 months, seeded anomalies and independent calculation truth.
32 benchmark cases	Executed deterministic baseline. An incomplete weighted score is shown honestly; it is not a live LLM ranking.

Next.js 15 · React 19 · TypeScript · PostgreSQL · Python · Power BI Embedded

INTELLIGENCE OS / 01

Architecture that can be inspected

The application preserves the original Axiom template and adds a persistent server-backed analytics workspace.



Server-only providers: OpenAI planning / MSAL + Power BI token issuance

Provisioning identity is separate from the web, worker and analytical DB roles.

The security boundary is on the server

Authenticated requests resolve real workspace membership. PostgreSQL uses a non-owner runtime role, forced row-level security and transaction-local user/workspace context. A client workspace selector cannot grant access.

Queries	Catalog plans become parameterized SQL. An AST validator rejects writes, unsafe functions, extra statements and unapproved relations. Analytical execution uses a read-only role.
Imports + jobs	Generated storage identifiers, bounded parsing, immutable versions and a durable queue. Job execution rechecks membership and supports cancellation.
Provider secrets	OpenAI and Microsoft credentials stay server-side. Only a scoped, short-lived Power BI embed configuration reaches the browser.
Exports	CSV requests reauthorize the user and audit snapshot metadata. Power BI artifacts require their creator and an unchanged server-derived report/RLS identity scope.

Deployment boundary

Docker runs web and worker as non-root processes alongside PostgreSQL and optional local Keycloak. Shared-volume artifacts make this a single-host reference deployment; scale-out needs an object-store adapter, production identity and operational backups.

INTELLIGENCE OS / 02

One workspace, connected journeys

Use the persistent shell and command palette to move from a metric to its evidence, then save a reproducible view.

Executive overview	Filter dates, plan and region; compare periods; inspect definitions; switch between chart and table; save views and export the displayed snapshot.
Data workbench	Import CSV, XLSX, JSON or Parquet. Inspect schema and quality issues, preview corrections, then publish a new immutable version.
AI Analyst	Ask a question against an authorized source. Inspect calculated facts, query, filters, version and limitations. Follow-ups preserve context; ambiguous requests trigger clarification.
Analysis Lab	Cohorts, observed funnels, contribution, robust anomaly detection, forecast backtests/intervals, comparisons and editable scenarios use deterministic calculations.
Catalog + lineage	Trace metric definitions and dataset transformations to analyses and dashboards. Inspect exact versions and explicit unavailable dependencies.
Evaluation center	Run known-answer cases and inspect correctness, provenance, safety, uncertainty and resource evidence. Compare stored model/prompt runs without fabricating missing scores.
Power BI + governance	Open approved reports, inspect refresh status and export options; review saved work, jobs, audit events and analyst/viewer/admin/owner roles.

Suggested first session

Filter the Executive overview → inspect a definition → import a small CSV → preview and publish a correction → ask for an aggregate → inspect its query and source version → save the analysis → run the evaluation baseline.

The interface distinguishes missing, zero, partial and stale results. Automated checks cover keyboard navigation and representative responsive layouts; manual accessibility review remains necessary.

Demo and connected data

Mode selection is explicit. A failed live service never silently becomes a successful sample response.

Demo	Seeded SaaS facts and a labeled deterministic question planner. Database permissions, jobs, ingestion and numerical calculations are real. Power BI is a local preview.
Connected	Verified identity and configured providers. Operational dashboards use a reviewed canonical snapshot; uploaded datasets remain a separate analysis source. Missing configuration produces a clear error.

Publish a canonical operational snapshot

```
npm run facts:import -- WORKSPACE_ID VERIFIED_ACTOR_ID
  /secure/path/snapshot.json
```

The command is one line; the wrap above is for readability. Run it in an isolated operator environment with the migration credential. The actor must already be an administrator or owner of the target workspace.

Four fact grains	Invoices: amount_cents. Infrastructure: cost_cents. Usage: integer tokens. Subscriptions: one customer snapshot per month, on the final UTC calendar day.
Explicit contracts	USD integer cents, UTC dates and token units. Fixed plan/region mappings; no implicit FX or unit conversion. Include explicit zero-MRR snapshots for known churn.
Versions + freshness	Sorted normalized content creates a SHA-256 version. Publication and activation are atomic; old versions remain available. Upstream refresh time and interval determine stale status.
Bounds	Canonical JSON: 50 MiB / 250,000 facts. Uploaded tables: 50 MiB file limit and decoded row/column limits. Query time, result size and subprocess resources are bounded.

Vendor extraction and business mapping remain your upstream pipeline's responsibility. Source coverage is declared and reviewed by the operator; schema validation cannot establish upstream completeness. See docs/DATA-SOURCES.md for a complete JSON example and recovery steps.

INTELLIGENCE OS / 04

Connect Power BI deliberately

App-owns-data integration is implemented with the Microsoft SDK and server-issued embed tokens. Live Microsoft verification still needs your tenant and capacity.

1 · Provision identity	Register an Entra application/service principal. Restrict the Power BI tenant setting to an approved security group and grant the required workspace/profile access.
2 · Assign capacity	Use eligible production A/F capacity or a supported existing EM/P arrangement. PPU alone is insufficient for production app-owns-data embedding and exportToFile.
3 · Build the model	Use the four fact tables and Date/Customer/Plan/Region dimensions. Configure single-direction dimension-to-fact relationships, explicit DAX measures and a verified dynamic RLS role.
4 · Map each tenant	Use a separate Power BI workspace/profile per Axiom organization. Install approved report/model IDs and application-user effective identities with the operator mapping command.
5 · Validate access	Test restricted and unrestricted identities, cross-tenant denial, revoked access, token renewal, report errors, filter synchronization and inactive-tab recovery.

```
npm run powerbi:export -- atlas
npm run powerbi:configure -- atlas .runtime/atlas-powerbi.json
```

What is supplied

A server adapter, SDK viewer, page/bookmark/filter controls, renewal, queued exports and refresh submission/history; sample table exporter, relationship instructions and DAX catalog. No generated or validated PBIX/PBIP artifact is claimed.

What remains live-only

Actual Microsoft authentication, capacity eligibility, semantic-model RLS, embed interactions, export contents and refresh completion remain unverified. Refresh submission is not completion. Check gateway credentials, tenant settings and licensing before enabling those actions.

Detailed setup and current official Microsoft reference links: docs/POWER-BI.md. Never use Publish to web for private business data.

INTELLIGENCE OS / 05

Run it. Verify it. Connect it.

Local reference evidence recorded on 23 October 2026. The repository documentation remains the detailed engineering source of truth.

```
npm ci
npm run setup
docker compose up --build -d
```

Open http://localhost:3000/login/. Setup generates ignored local credentials. Initialization applies migrations and seeds the reference workspaces. Local demo sign-in and Keycloak start-dev are development facilities.

Domain + parsers	23 unit tests passed; 8 Python analytical tests passed.
Persistent integration	21 integration tests passed, including the 32-case deterministic benchmark baseline. Missing model-cost evidence keeps the weighted benchmark score incomplete.
Browser journeys	5 production-browser tests passed. Eight workspace routes had no detected axe A/AA violations; tested widths were 320, 1440 and 3840 CSS pixels.
Build + packaging	Typecheck, production build and Docker image builds passed. Fresh reference-stack runtime smoke also passed.
External services	Local Keycloak OIDC was exercised. Live OpenAI model evaluation and live Power BI checks remain unverified without configured services.

Reproduce the checks

```
npm run typecheck && npm test
.venv/bin/python python/test_analytics.py
npm run test:integration
npm run build && npm run start
# Another terminal; worker + database must also run:
npm run test:e2e
```

These results do not establish WCAG certification, production capacity or sub-500 ms loading. See docs/VALIDATION.md, docs/PERFORMANCE.md and docs/FEATURE-STATUS.md for exact scope and remaining work.

Open the feature branch and full engineering handoff